CYBER SECURITY

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🌐 Understanding TCP/IP Model for Cyber Security

- TCP/IP Model = 4 Layers → Application, Transport, Internet, Network Access
- Each layer has a specific role & different cyber attacks target each layer.

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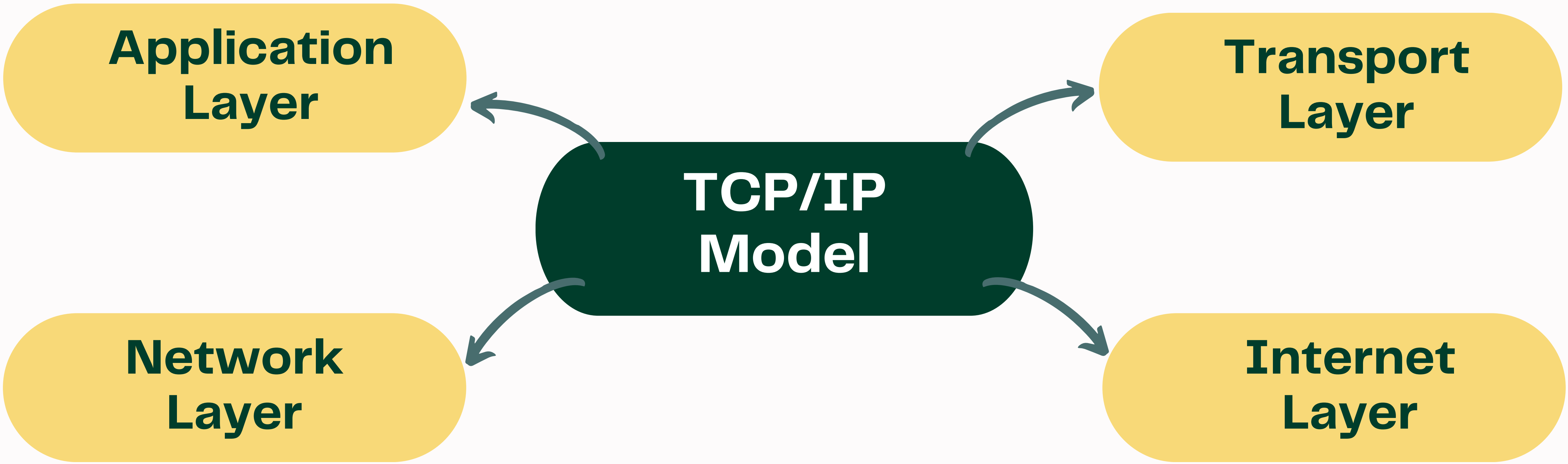

Application Layer (👨‍💻 User Interface)

- Provides services to users (HTTP, HTTPS, DNS, FTP, SMTP).
- Phishing via fake HTTP sites.
- DNS Spoofing → redirects to fake IPs.
- Malware through email (SMTP abuse).
- User types google.com, DNS poisoning can redirect to a fake site.

Transport Layer (🚚 Reliable Delivery)

- Breaks data into segments; ensures delivery via TCP (reliable) or UDP (fast).
- Port Scanning (e.g., with Nmap).
- Session Hijacking.
- YouTube streaming uses UDP; attacker sniffing the same port can monitor activity.

Internet Layer
(🌍 Logical Addressing)

- Handles IP addressing & routing (IP, ICMP).
- IP Spoofing (fake source IP).
- DDoS Attacks (botnets flooding IPs).
- A website taken down due to DDoS traffic at Internet Layer.

Network Access Layer (📡 Physical + Data Link)

- Deals with hardware & physical transmission (Ethernet, Wi-Fi, MAC).
- MAC Spoofing.
- Packet Sniffing (Wireshark).
- Rogue Access Points (fake Wi-Fi).
- Free coffee shop Wi-Fi with fake hotspot → user traffic gets sniffed.